

ABHISHEK SUBRAMANIAN

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EDUCATION

Virginia Tech

Master of Science in Computer Engineering - AI/ML and SWE concentration (GPA 3.7 / 4.0)

Blacksburg, VA

Aug 2024 – Dec 2025

- **Relevant Courses:** Advanced Machine Learning, Computer Vision, Natural Language Processing, Advanced Technological Singularity, Applications of Machine Learning, Large Scale Software Systems Engineering

College of Engineering, Guindy (Anna University)

B.E. Electronics and Communications Engineering (GPA 8.01 / 10.00)

Chennai, India

Aug 2018 – Apr 2022

- **Relevant Courses:** Data structures and Object Oriented Programming, Operating Systems, Soft Computing

PROFESSIONAL EXPERIENCE

Quantiphi

Bengaluru, India

Full-Stack Software Developer

Aug 2022 – May 2024

- Built a responsive frontend using **React.js, Material UI, and Figma** to interact with NVIDIA's Digital Avatar, powered by Llama's 33B LLM, incorporating a RAG pipeline to contextually accurate responses and improving conversation quality by **30%**.
- Optimized asynchronous data flow between the frontend and backend to reduce latency in dynamic UI rendering, achieving a **40% reduction** in load time and enhancing user responsiveness.
- Integrated NVIDIA Riva's ASR and TTS APIs to enable **real-time captions and chat functionality**, improving accessibility and **increasing user engagement by 35%**.
- Led the **backend development** for a project entailing intelligent **music synthesis using machine learning**, implementing **authentication systems, music configurators, and server monitoring tools**.
- Devised a secure **MySQL database** that supports **100+ concurrent users** and **optimized data handling through multithreading and parallel computing**, developing **scalable Flask APIs** with **CORS and WebSockets** for real-time frontend communication.
- Containerized the entire application stack using **Docker**, enhancing scalability and maintainability and **reducing deployment time by 40%**.
- Built a full-stack **Generative AI-powered Digital Avatar chatbot** using **React.js (frontend)** and **Express.js (backend)**, leveraging the **Tokio framework** to deliver real-time personalized recommendations, enhancing user experience and **boosting interaction metrics by 25%**.
- Engineered **gRPC API calls** to enable real-time processing and communication between NVIDIA Riva & Tokio microservices and text-based input systems, reducing cross-service latency by **~200ms** and ensuring seamless message synchronization.

SKILLS

Professional Certifications: Google Cloud Certified: Associate Cloud Engineer  and Professional Cloud Developer 

Programming Languages: Python, JavaScript, TypeScript, SQL, C++, C#, R, Java, HTML, CSS, Matlab, Ruby, GoLang.

Frameworks and Tools: Flask, Node.js, MySQL, PyTorch, TensorFlow, Git, Linux, NoSQL Firebase, React.js, Redux, Material UI, Bootstrap, Figma, Pygame, paho-mqtt, Pandas, FastAPI, PySpark, Android Studio, Hadoop, CUDA, LangChain, LangGraph.

Cloud and DevOps: Google Cloud Platform (GCP), NVIDIA Avatar Cloud Engine (ACE), Docker, Kubernetes, Jenkins, Bazel, AWS.

SELECTED PROJECTS

Explainable AI for Analyzing Transformer Models in NLP Tasks

Jan 2025 – Present

- Interpreted transformer model decisions across **NLP** tasks using **SHAP** to enhance transparency and cross-model comparison.
- Fine-tuned **BERT, RoBERTa**, and **T5** on **sentiment analysis, classification, NER**, and **QA**, extracting token-level attributions with SHAP.
- Analyzed model behaviors via heatmaps and dependency plots to expose biases and explainability gaps.

Enhancing Emotional Well-Being through ML based Music Emotion Recognition

Sep 2024 - Dec 2024

- **Designed and fine-tuned CNN and RNN (LSTM) models** using MFCC-based features to capture temporal audio patterns, achieving **76.22% accuracy** on arousal-valence predictions using the DEAM dataset.
- **Validated emotion recognition performance** for mental health applications, demonstrating strong model generalization for therapeutic use cases.

YOLOv5-Powered X-Ray Baggage Screening for Threat Detection in Airports

Sept 2024 – Dec 2024

- Created a custom YOLOv5 model using transfer learning with **CSPDarknet53**, fine-tuned on the OPIXray dataset to detect and classify suspicious items in airport baggage scans.
- Boosted precision (**90.35%**) and recall (**87.16%**) by resolving class-specific performance issues using dataset augmentation, dynamic resizing, and anchor box optimization.

Unhealthy Liver Detection using CNN with IoT

Feb 2023 – Jun 2023

- Presented a CNN-based method for liver disease detection using CT images at the **2023 IEEE ICSCSS** in Coimbatore, India, demonstrating the model's potential for early liver disease diagnosis.
- Applied advanced image processing techniques such as augmentation, enhancement, and restoration to improve model training and dataset variability, achieving a **30% increase in accuracy and an 86.8% detection rate**.